



```
[8] ✓ 0s df = df[['text', 'label']]

print(df.shape)
df.head()
```

... (44898, 2)

	text	label
0	Donald Trump just couldn t wish all Americans ...	0
1	House Intelligence Committee Chairman Devin Nu...	0
2	On Friday, it was revealed that former Milwauk...	0
3	On Christmas day, Donald Trump announced that ...	0
4	Pope Francis used his annual Christmas Day mes...	0



Next steps:

Generate code with df

New interactive sheet

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[9]

✓ 0s

 `print(df.isnull().sum())`

...
text 0
label 0
dtype: int64

[10]

✓ 0s

 `df = df.sample(frac=1, random_state=42).reset_index(drop=True)`
`df.head()`

...

	text	label
0	21st Century Wire says Ben Stein, reputable pr...	0
1	WASHINGTON (Reuters) - U.S. President Donald T...	1
2	(Reuters) - Puerto Rico Governor Ricardo Rosse...	1
3	On Monday, Donald Trump once again embarrassed...	0
4	GLASGOW, Scotland (Reuters) - Most U.S. presid...	1

Next steps: [Generate code with df](#) [New interactive sheet](#)

[]

Start coding or [generate](#) with AI.

```
[20] model = Sequential([
      Embedding(input_dim=5000, output_dim=64, input_length=200),
      LSTM(64),
      Dense(1, activation='sigmoid')
    ])

model.compile(
    loss='binary_crossentropy',
    optimizer='adam',
    metrics=['accuracy']
)

model.summary()
```

```
... /usr/local/lib/python3.12/dist-packages/keras/src/layers/core/embedding.py:100: UserWarning: Argument `input_length` is deprecated. Just remove it.
      warnings.warn(
Model: "sequential"
```

Layer (type)	Output Shape	Param #
embedding (Embedding)	?	0 (unbuilt)
lstm (LSTM)	?	0 (unbuilt)
dense (Dense)	?	0 (unbuilt)

Total params: 0 (0.00 B)
Trainable params: 0 (0.00 B)
Non-trainable params: 0 (0.00 B)



[21]
✓ 1m

```
history = model.fit(  
    X_train_pad,  
    y_train,  
    epochs=3,  
    batch_size=32,  
    validation_split=0.2  
)
```

```
... Epoch 1/3  
200/200 _____ 21s 94ms/step - accuracy: 0.9261 - loss: 0.2095 - val_accuracy: 0.9669 - val_loss: 0.0858  
Epoch 2/3  
200/200 _____ 20s 100ms/step - accuracy: 0.9880 - loss: 0.0398 - val_accuracy: 0.9756 - val_loss: 0.0870  
Epoch 3/3  
200/200 _____ 19s 95ms/step - accuracy: 0.9922 - loss: 0.0285 - val_accuracy: 0.9844 - val_loss: 0.0513
```

```
[23] ✓ 0s  
plt.figure(figsize=(8,5))  
plt.plot(history.history['accuracy'], label='Training Accuracy')  
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')  
  
plt.title('Model Accuracy')  
plt.xlabel('Epoch')  
plt.ylabel('Accuracy')  
plt.legend()  
  
plt.show()
```

